Knowledge-Based Systems 000 (2016) 1–12

Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys

CC-integrals: Choquet-like Copula-based aggregation functions and its application in fuzzy rule-based classification systems

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ARTICLE INFO

Article history:
Received 3 May 2016
Revised 21 October 2016
Accepted 1 December 2016
Available online xxx

Keywords:
Aggregation functions
Choquet integral
Copula
t-norm
Overlap function
Fuzzy rule-based classification systems
Fuzzy reasoning method

ABSTRACT

This paper introduces the concept of Choquet-like Copula-based aggregation function (CC-integral) and its application in fuzzy rule-based classification systems. The standard Choquet integral is expanded by distributing the product operation. Then, the product operation is generalized by a copula. Unlike the generalization of the Choquet integral by t-norms using its standard form (i.e., without distributing the product operator), which results in a pre-aggregation function, the CC-integral satisfies all the conditions required for an aggregation function. We build some examples of CC-integrals considering different examples of copulas, including t-norms, overlap functions and copulas that are neither t-norms nor overlap functions. We show that the CC-integral based on the minimum t-norm, when applied in fuzzy rule-based classification systems, obtains a performance that is, with a high level of confidence, better than that which adopts the winning rule (maximum). We concluded that the behavior of CC-integral is similar to the best Choquet-like pre-aggregation function. Consequently, the CC-integrals introduced in this paper can enlarge the scope of the applications by offering new possibilities for defining fuzzy reasoning methods with a similar gain in performance.

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1. Introduction

Whenever we face the task of organizing instances in various predefined categories according to their different features, we have the so-called classification problem [1]. In general, one applies a learning algorithm, which, using the available information about the instances, is able to learn a decision function, called classifier, which is used to perform the classification for future instances.

Classification problems are present in many different real-world problems. For example, Patidar et al. [2] developed a method for diagnosis of coronary artery disease classifying heart rate signals, Galar et al. [3,4] presented a survey in fingerprint classification, Huang and Lin [5] developed a system to classify multiple harmonic sources (which distort the original frequency) in a power quality problem and finally, a study using different classifiers in earthquake predictions is done in [6].

Fuzzy Rule-Based Classification Systems (FRBCSs) [7] are widely used to deal with classification problems, since they usually present a good performance besides providing an interpretable model. The usage of linguistic terms for modeling the problem domain allows these systems to be easily applied and understood by the final users of real world applications [8,9]. Moreover, FRBCSs can combine information coming from different sources, that is, expert knowledge, mathematical models, data bases, or empirical measures [9].

An important issue in any FRBCS is the considered Fuzzy Reasoning Method (FRM) [10], which consists in an inference procedure that uses the information stored in the knowledge base to determine the class in which new instances will be classified. To do so, in first place, one computes the compatibility between the example to be classified and each rule, which provides local information. Then, this local information is aggregated, obtaining the global information associated with each known class. Finally, the class determined for the new example is the one presenting the maximum global information.

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http://dx.doi.org/10.1016/j.knosys.2016.12.004
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The FRM of the Winning Rule (WR), which uses the maximum as aggregation function to obtain the global information, is widely used in the literature [11]. However, this method considers, for each class, just the information given by the fuzzy rule with the highest compatibility with the example and, thus, the available information provided by all the other fired fuzzy rules is ignored. In order to avoid this shortcoming using averaging operators, Barrenechea et al. [12] introduced a FRM that aggregates the information given by all the fired fuzzy rules using the Choquet integral [13]. Specifically, they achieved the best results when using the Choquet integral considering the power measure and applying the CHC evolutionary algorithm [14] to learn a different fuzzy measure for each class of the problem.

Aimed at improving the results in [12], Lucca et al. [15] introduced the concept of pre-aggregation function. These functions satisfy the same boundary conditions of an aggregation function, but are just directionally increasing [16], that is, they are increasing along some specific ray (direction), but not for all directions. Furthermore, they presented the Choquet-like construction method of pre-aggregation functions, which are built by replacing the product operation of the Choquet integral by other t-norms. Finally, they applied these Choquet-like pre-aggregation functions in the FRM of the FRBCSs, showing that the use of Hamacher Product t-norm along with the power measure, outperformed the performance of the method by Barrenechea et al. [12] as well as that of the WR.

From the methods introduced in [12,15], the questions that arise are: "Is there a way to build a Choquet-like integral that is based on a more general aggregation than the product operator, but that yields to an aggregation function instead of a pre-aggregation function? If so, is the FRBCS using this new method able to achieve a better or a similar performance than that of the most accurate Choquet-like pre-aggregation function?"

To answer the questions above, the objectives of this paper are:

1. To introduce the notion of Choquet-like Copula-based aggregation functions, called CC-integrals;
2. To define different types of CC-integrals, using t-norms [17], overlap functions [18–21], and copulas [22] that are neither t-norms nor overlap functions;
3. To analyze the behavior of CC-integrals when applied in FRBCSs.

To test the quality of the FRBCS in which the FRM uses CC-integrals we consider 30 real-world problems that are publicly available in the KEEL database repository [23]. The performance of the classifiers is measured using the standard accuracy rate and the results are supported by appropriate statistical tests [24–26]. Specifically, we analyze the behavior of the different CC-integrals considered in this paper and we test if the best performing one is competitive versus the FRM of the WR, the classical Choquet integral and the best Choquet-like based pre-aggregation function, that is, the one using the Hamacher t-norm (with the power measure[1]), which is denoted by Ham_{PA} in this paper.

The paper is organized as follows. Section 2 presents some basic concepts that are necessary to develop the paper. Section 3 introduces the concept of CC-integral. The FRBCS using CC-integrals is presented in Section 4. In Section 5, we explain the experimental framework, including the description of the 30 datasets considered in this paper and statistical tests, used for comparing the achieved results. The analysis of the application and run-time of the CC-Integral in classification problems are presented in Section 6 and Section 7 is the Conclusion.

In order to ease the readability of the paper, we have created three appendixes: the first one (Appendix A) containing a summary of the abbreviations used in this paper, the second (Appendix B) in which is present an example of an overlap function that is a copula and finally in the last one (Appendix C), where is shown the proof of the CC-integral is an increasing function.

2. Preliminary concepts

This section aims at introducing the background necessary to develop the paper. First, we present some basic concepts, and then some new results that are important in the context of the paper.

2.1. Basic concepts

The key concept in this paper is one of aggregation functions [11,27]:

Definition 1. A function A: $[0, 1]^n \to [0, 1]$ is said to be an aggregation function whenever the following conditions are satisfied:

(A1) A is increasing[2] in each argument: for each $i \in \{1, \ldots, n\}$, if $x_i \leq y$, then $A(x_1, \ldots, x_n) \leq A(x_1, \ldots, x_{i-1}, y, x_{i+1}, \ldots, x_n)$;
(A2) A satisfies the boundary conditions: $A(0, \ldots, 0) = 0$ and $A(1, \ldots, 1) = 1$.

Definition 2. An aggregation function T: $[0, 1]^2 \to [0, 1]$ is a t-norm if, for all $x, y, z \in [0, 1]$, it satisfies the following properties:

(T1) Commutativity: $T(x, y) = T(y, x)$;
(T2) Associativity: $T(x, T(y, z)) = T(T(x, y), z)$;
(T3) Boundary condition: $T(x, 1) = x$.

If T satisfies ***(T3)*** (and also $T(1, x) = x$ only), then it is called a semi-copula.

Definition 3. A function O: $[0, 1]^2 \to [0, 1]$ is said to be an overlap function if it satisfies the following conditions:

(O1) O is commutative;
(O2) $O(x, y) = 0$ if and only if $xy = 0$;
(O3) $O(x, y) = 1$ if and only if $xy = 1$;
(O4) O is increasing;
(O5) O is continuous.

Definition 4. A bivariate function C: $[0, 1]^2 \to [0, 1]$ is a copula if it satisfies the following conditions, for all $x, x', y, y' \in [0, 1]$ with $x \leq x'$ and $y \leq y'$:

(C1) $C(x, y) + C(x', y') \geq C(x, y') + C(x', y)$;
(C2) $C(x, 0) = C(0, x) = 0$;
(C3) $C(x, 1) = C(1, x) = x$.

Copulas are functions that link (two-dimensional) probability distribution functions to their one-dimensional margins, playing an important role in the theory of probabilistic metric spaces and statistics [22].

Proposition 1 [17, Proposition 9.8] [28, Lemma 6.1.8, Lemma 6.3.1]**.** *For each copula C: $[0, 1]^2 \to [0, 1]$, the Lukasiewicz and Minimum T-norms (T_L, T_M) T: $[0, 1]^2 \to [0, 1]$ it holds that:*

(i) $T_L \leq C \leq T_M$;
(ii) *C is increasing;*
(iii) *C satisfies the Lipschitz property with constant 1, that is, for all $x_1, x_2, y_1, y_2 \in [0, 1]$, one has that:*

$$| C(x_1, y_1) - C(x_2, y_2) | \leq | x_1 - x_2 | + | y_1 - y_2 | .$$

[1] For more explanation of the power measure see, [12,15].

[2] For an increasing (decreasing) function we do not mean a strictly increasing (decreasing) function.

Please cite this article as: G. Lucca et al., CC-integrals: Choquet-like Copula-based aggregation functions and its application in fuzzy rule-based classification systems, Knowledge-Based Systems (2016), http://dx.doi.org/10.1016/j.knosys.2016.12.004

Table 1
Examples of Copulas.

(I) T-norms		
Definition	Name	Observations
$T_M(x,y) = \min\{x,y\}$	Minimum	Overlap function
$T_P(x,y) = xy$	Algebraic Product	Overlap function
$T_L(x,y) = \max\{0, x+y-1\}$	Lukasiewicz	
$T_{HP}(x,y) = \begin{cases} 0 & \text{if } x=y=0 \\ \frac{xy}{x+y-xy} & \text{otherwise} \end{cases}$	Hamacher Product	Overlap function
(II) Non-associative overlap functions		
Definition	Reference	Observations
$O_B(x,y) = \min\{x\sqrt{y}, y\sqrt{x}\}$	[18, Theorem 8]	Cuadras-Augé family of copulas [32]
$O_{mM}(x,y) = \min\{x,y\}\max\{x^2,y^2\}$	[33, Example 3.1.(i)], [34, Example 4] [35, Example 3.1]	
$O_\alpha(x,y) = xy(1+\alpha(1-x)(1-y)),\ \alpha \in [-1,0[\ \cup\ ]0,1]$	[22, Apendix A (A.2.1)], [36]	
(III) Other non-associative copulas		
Definition	Reference	Observations
$C_F(x,y) = xy + x^2y(1-x)(1-y)$	[17, Example 9.5 (v)]	Non-commutative
$C_L(x,y) = \max\{\min\{x, \frac{y}{2}\}, x+y-1\}$	[22, Apendix A (A.5.3a)]	Non-commutative
$C_{Div}(x,y) = \frac{xy+\min\{x,y\}}{2}$	[22, Apendix A (A.8.7)]	

An immediate consequence of Proposition 1 is that any copula is continuous. Then, each associative copula is a continuous t-norm [17, Corollary 9.9].

Theorem 1 [17, Theorem 9.10]**.** *Let T: $[0,1]^2 \to [0,1]$ be a t-norm. Then, the following statements are equivalent:*

(i) *T is copula.*
(ii) *T satisfies the Lipschitz property with constant 1.*

The definitions of overlap, t-norms and copulas can be easily extended to n-ary functions. [17,29–31].

Table 1 presents the copulas used in the rest of the paper, in particular, some t-norms T: $[0,1]^2 \to [0,1]$ (Table 1 (I)), overlap functions O: $[0,1]^2 \to [0,1]$ (Table 1 (II)) and copulas C: $[0,1]^2 \to [0,1]$ that are neither t-norms nor overlap functions (Table 1 (III)). Observe that the overlap functions and copulas shown in Table 1 are all non-associative.

Now, we present the concept of fuzzy measure [13,37], which is a central tool for defining the Choquet integral. In what follows, denote $N = \{1,\ldots,n\}$, for an arbitrary $n > 0$.

Definition 5. A function $\mathfrak{m} : 2^N \to [0,1]$ is said to be a fuzzy measure if, for all $X, Y \subseteq N$, it satisfies the following properties:

($\mathfrak{m}$1) Increasing: if $X \subseteq Y$, then $\mathfrak{m}(X) \leq \mathfrak{m}(Y)$;
($\mathfrak{m}$2) Boundary conditions: $\mathfrak{m}(\emptyset) = 0$ and $\mathfrak{m}(N) = 1$.

Regarding aggregation functions, we use fuzzy measures to analyze the relationship among the elements that we are aggregating, obtaining the relevance of a coalition. In this paper, we adopt the power measure $\mathfrak{m}_{PM} : 2^N \to [0,1]$, which is defined, for all $X \subseteq N$, by

$$\mathfrak{m}_{PM}(X) = \left(\frac{|X|}{n}\right)^q, \text{ with } q > 0. \tag{1}$$

The choice for this fuzzy measure was based on the results obtained by Barrenechea et al. [12], who introduced an evolutionary algorithm to define the most suitable q to be used in the definition of the measure for each class. See also the results shown in [15,36,38], which also make use of such approach.

The Choquet integral generalizes the Lebesgue integral, defined considering additive measures. The Choquet integral, however, considers fuzzy measures. The discrete Choquet integral [13] is defined on finite spaces:

Definition 6 [11, Definition 1.74]**.** Let $\mathfrak{m} : 2^N \to [0,1]$ be a fuzzy measure. The discrete Choquet integral is the function $\mathfrak{C}_\mathfrak{m} : [0,1]^n \to [0,1]$, defined, for all of $\vec{x} = (x_1,\ldots,x_n) \in [0,1]^n$, by:

$$\mathfrak{C}_\mathfrak{m}(\vec{x}) = \sum_{i=1}^{n} \left(x_{(i)} - x_{(i-1)}\right) \cdot \mathfrak{m}\left(A_{(i)}\right), \tag{2}$$

where $(x_{(1)},\ldots,x_{(n)})$ is an increasing permutation on the input x, that is, $0 \leq x_{(1)} \leq \ldots \leq x_{(n)}$, where $x_{(0)} = 0$ and $A_{(i)} = \{(i),\ldots,(n)\}$ is the subset of indices corresponding to the $n-i+1$ largest components of $\vec{x}$.

Observe that the Eq. (2) can be also written as:

$$\mathfrak{C}_\mathfrak{m}(\vec{x}) = \sum_{i=1}^{n} \left(x_{(i)} \cdot \mathfrak{m}\left(A_{(i)}\right) - x_{(i-1)} \cdot \mathfrak{m}\left(A_{(i)}\right)\right), \tag{3}$$

which we call the Choquet integral in its expanded form.

When using the Choquet integral in order to aggregate the inputs, it is possible to consider the relevance of the different coalitions (groups of inputs).

2.2. Some new results on copulas and overlap functions

Theorem 2. *A copula C: $[0,1]^2 \to [0,1]$ is an overlap function if it is commutative and positive.*

Proof. Is immediate that any positive and commutative copula C satisfies **(O1)** and **(O2)**. It follows that:

(O3): From **(C3)**, we have that $C(1,y) = 1$ if and only if $y = 1$. From the monotonicity of C, this implies that $C(x,y) = 1$ if and only if $x = y = 1$. Thus, one concludes that C satisfies **(O3)**.
(O4): From Proposition 1 **(ii)**, it follows that C is increasing.
(O5): It follows from Proposition 1 **(iii)** that, since C satisfies the Lipschitz condition for a constant 1, then C is continuous. □

Notice that any overlap function which is also a copula, by **(C3)**, necessarily has 1 as neutral element and, by Proposition 1, it satisfies the Lipschitz property with constant 1.

Theorem 3. *Let O: $[0,1]^2 \to [0,1]$ be an overlap function with neutral element and satisfying the Lipschitz condition for the constant 1.*

O is a copula if, for all x, y, x′, y′ ∈ [0, 1] with x ≤ x′ and y ≤ y′, there exists z ≤ z′ such that

$$O(O(x, y'), z') = O(x, y) \tag{4}$$

and

$$O(O(x', y'), z) = O(x', y). \tag{5}$$

Proof. Trivially, **(C2)** and **(C3)** follows from **(O2)** and the well-known fact that whenever an overlap function has a neutral element then this element is 1. To prove **(C1)**, consider $x, x', y, y', z, z' \in [0, 1]$ such that $x \le x'$, $y \le y'$ and the Eqs. (4) and (5) hold. It follows that:

$$\begin{aligned} O(x', y') - O(x, y') &\ge O\big(O(x', y'), z'\big) \\ &\quad - O\big(O(x, y'), z'\big) \text{ Lipschitz cond. for 1} \\ &= O\big(O(x', y'), z'\big) - O(x, y) \text{ by Eq. (4)} \\ &\ge O\big(O(x', y'), z\big) - O(x, y) \\ &= O(x', y) - O(x, y) \text{ by Eq. (5)} \end{aligned}$$

and then $O(x, y) + O(x', y') \ge O(x, y') + O(x', y)$. □

Observe that when O has a neutral element, by its continuity and isotonicity, there always exist $z, z' \in [0, 1]$ satisfying the Eqs. (4) and (5). An example of an overlap function that is also a copula function, is introduced in Appendix B.

3. Constructing Choquet-like Copula-based aggregation functions

In this section, we introduce a method for constructing a family of aggregation functions obtained by combining the discrete Choquet Integral in its expanded form (Eq. (3)) with copulas, just substituting the product operation in Eq. (3) by a copula. Such functions are called CC-integrals.

In the following, consider $N = \{1, \ldots, n\}$.

Definition 7. Let $\mathfrak{m} : 2^N \to [0, 1]$ be a fuzzy measure and C: $[0, 1]^2 \to [0, 1]$ be a bivariate copula. The Choquet-like Copula-based integral with respect to $\mathfrak{m}$ is defined as a function $\mathfrak{C}^C_{\mathfrak{m}} : [0, 1]^n \to [0, 1]$, given, for all $x \in [0, 1]^n$, by

$$\mathfrak{C}^C_{\mathfrak{m}}(\vec{x}) = \sum_{i=1}^{n} C\big(x_{(i)}, \mathfrak{m}\big(A_{(i)}\big)\big) - C\big(x_{(i-1)}, \mathfrak{m}\big(A_{(i)}\big)\big), \tag{6}$$

where $(x_{(1)}, \ldots, x_{(n)})$ is an increasing permutation on the input x, that is, $0 \le x_{(1)} \le \ldots \le x_{(n)}$, with the convention that $x_{(0)} = 0$, and $A_{(i)} = \{(i), \ldots, (n)\}$ is the subset of indices of $n - i + 1$ largest components of $\vec{x}$.

Proposition 2. *Under the conditions of Definition 7, $\mathfrak{C}^C_{\mathfrak{m}}$ is well defined, for any copula C: $[0, 1]^2 \to [0, 1]$.*

Proof. Since $x_{(i)} \ge x_{(i-1)}$ and, by Proposition 1, C is increasing, then it is immediate that $\mathfrak{C}^C_{\mathfrak{m}}(\vec{x}) \ge 0$, for any $\vec{x} \in [0, 1]$. On the other hand, by Proposition 1, C satisfies the Lipschitz property with constant 1, that is, for all $x_1, x_2, y_1, y_2 \in [0, 1]$, one has that:

$$| C(x_1, y_1) - C(x_2, y_2) | \le | x_1 - x_2 | + | y_1 - y_2 | .$$

Then, since $x_{(i)} - x_{(i-1)} \ge 0$ and C is increasing, one has that:

$$\begin{aligned} C(x_{(i)}, \mathfrak{m}(A_{(i)})) - C(x_{(i-1)}, \mathfrak{m}(A_{(i)})) &\le x_{(i)} - x_{(i-1)} + \mathfrak{m}(A_{(i)}) - \mathfrak{m}(A_{(i)}) \\ &= x_{(i)} - x_{(i-1)}. \end{aligned}$$

Thus, for any $\vec{x} \in [0, 1]$, it follows that,

$$\begin{aligned} \mathfrak{C}^C_{\mathfrak{m}}(\vec{x}) &= \sum_{i=1}^{n} C\big(x_{(i)}, \mathfrak{m}\big(A_{(i)}\big)\big) - C\big(x_{(i-1)}, \mathfrak{m}\big(A_{(i)}\big)\big) \\ &\le \sum_{i=1}^{n} x_{(i)} - x_{(i-1)} \\ &= x_{(n)} \\ &\le 1. \end{aligned}$$

□

Consider a fuzzy measure $\mathfrak{m} : 2^N \to [0, 1]$ and $\vec{x} \in [0, 1]^n$. The CC-integrals with respect to $\mathfrak{m}$, for each copula of Table 1, assume a form included in Table 2.

Proposition 3. *For any copula C: $[0, 1]^2 \to [0, 1]$ and fuzzy measure $\mathfrak{m} : 2^N \to [0, 1]$, $\mathfrak{C}^C_{\mathfrak{m}}$ is idempotent.*

Proof. Considering $\vec{x} = (x, \ldots, x) \in [0, 1]^n$, one has that:

$$\begin{aligned} \mathfrak{C}^C_{\mathfrak{m}}(\vec{x}) &= C(x, \mathfrak{m}(A_{(1)})) - C(0, \mathfrak{m}(A_{(1)})) \\ &\quad + \sum_{i=2}^{n} C(x, \mathfrak{m}(A_{(i)})) - C(x, \mathfrak{m}(A_{(i)})) \\ &= C(x, 1) - C(0, 1) + 0 \\ &= x. \end{aligned}$$

□

Proposition 4. *For any copula C: $[0, 1]^2 \to [0, 1]$ and fuzzy measure $\mathfrak{m} : 2^N \to [0, 1]$, $\mathfrak{C}^C_{\mathfrak{m}}$ satisfies the boundary conditions* **(A2)**.

Proof. Considering $\vec{0} = (0, \ldots, 0) \in [0, 1]^n$ and $\vec{1} = (1, \ldots, 1) \in [0, 1]^n$, by Proposition 3, one has that $\mathfrak{C}^C_{\mathfrak{m}}(\vec{0}) = 0$ and $\mathfrak{C}^C_{\mathfrak{m}}(\vec{1}) = 1$. □

Proposition 5. *For any copula C: $[0, 1]^2 \to [0, 1]$ and fuzzy measure $\mathfrak{m} : 2^N \to [0, 1]$, $\mathfrak{C}^C_{\mathfrak{m}}$ is increasing* **(A1)**.

In order to ease the readability of the paper, the proof to the Proposition 5 was moved to Appendix C and from there, we can imply that the following results are immediate.

Corollary 1. *For any bivariate copula C: $[0, 1]^2 \to [0, 1]$ and fuzzy measure $\mathfrak{m} : 2^N \to [0, 1]$, it holds that* $\min \le \mathfrak{C}^C_{\mathfrak{m}} \le \max$.

Theorem 4. *For any bivariate copula C: $[0, 1]^2 \to [0, 1]$ and fuzzy measure $\mathfrak{m} : 2^N \to [0, 1]$, $\mathfrak{C}^C_{\mathfrak{m}}$ is an average aggregation function.*

Proof. It follows from Propositions 4 and 5, and Corollary 1. □

4. Applying CC-integrals as aggregation functions in a fuzzy reasoning algorithm

In this section, we recall firstly the main concepts of FRBCSs and, then, the algorithm of the proposed FRM using CC-integrals is presented.

4.1. Fuzzy rule-based classification systems

A classification problem is composed by m training examples characterized by n input attributes, denoted by

$$\vec{x}_p = (x_{p1}, \ldots, x_{pn}, y_p),$$

where

- $p = 1, \ldots, m$;
- x_{pi}, with $i = 1, \ldots, n$, is the value of the ith attribute;

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Table 2
Choquet-like Copula-based integral (CC-integral).

Copula	CC-integral
TM	$\mathfrak{C}_{\mathfrak{m}}^{C_M}(\vec{x}) = \sum_{i=1}^{n}(\min\{x_{(i)}, \mathfrak{m}(A_{(i)})\} - \min\{x_{(i-1)}, \mathfrak{m}(A_{(i)})\})$
TL	$\mathfrak{C}_{\mathfrak{m}}^{CL}(\vec{x}) = \sum_{i=1}^{n}(\max\{0, x_{(i)} + \mathfrak{m}(A_{(i)}) - 1\} - \max\{0, x_{(i-1)} + \mathfrak{m}(A_{(i)}) - 1\})$
TH	$\mathfrak{C}_{\mathfrak{m}}^{C_{HP}}(\vec{x}) = \begin{cases} 0, & \text{if } x = y = 0 \\ \frac{x_{(i)}\mathfrak{m}(A_{(i)})}{x_{(i)}+\mathfrak{m}(A_{(i)})-x_{(i)}\mathfrak{m}(A_{(i)})} - \frac{x_{(i-1)}\mathfrak{m}(A_{(i)})}{x_{(i-1)}+\mathfrak{m}(A_{(i)})-x_{(i-1)}\mathfrak{m}(A_{(i)})}, & \text{otherwise} \end{cases}$
OB	$\mathfrak{C}_{\mathfrak{m}}^{C_{O_B}}(\vec{x}) = \sum_{i=1}^{n}(\min\{x_{(i)}\sqrt{\mathfrak{m}(A_{(i)})}, \mathfrak{m}(A_{(i)})\sqrt{x_{(i)}}\} - \min\{x_{(i-1)}\sqrt{\mathfrak{m}(A_{(i)})}, \mathfrak{m}(A_{(i)})\sqrt{x_{(i-1)}}\})$
Oα	$\mathfrak{C}_{\mathfrak{m}}^{C_{O_\alpha}}(\vec{x}) = \sum_{i=1}^{n}(x_{(i)}\mathfrak{m}(A_{(i)})(1+\alpha(1-x_{(i)})(1-\mathfrak{m}(A_{(i)}))) - x_{(i-1)}\mathfrak{m}(A_{(i)})(1+\alpha(1-x_{(i-1)})(1-\mathfrak{m}(A_{(i)}))))$
OmM	$\mathfrak{C}_{\mathfrak{m}}^{C_{O_{mM}}}(\vec{x}) = \sum_{i=1}^{n}(\min\{x_{(i)}, \mathfrak{m}(A_{(i)})\}\max\{{x_{(i)}}^2, \mathfrak{m}(A_{(i)})^2\} - \min\{x_{(i-1)}, \mathfrak{m}(A_{(i)})\}\max\{{x_{(i-1)}}^2, \mathfrak{m}(A_{(i)})^2\})$
CF	$\mathfrak{C}_{\mathfrak{m}}^{C_F}(\vec{x}) = \sum_{i=1}^{n}(x_{(i)}\mathfrak{m}(A_{(i)}) + {x_{(i)}}^2\mathfrak{m}(A_{(i)})(1-x_{(i)})(1-\mathfrak{m}(A_{(i)})) - x_{(i-1)}\mathfrak{m}(A_{(i)}) + {x_{(i-1)}}^2\mathfrak{m}(A_{(i)})(1-x_{(i-1)})(1-\mathfrak{m}(A_{(i)})))$
CL	$\mathfrak{C}_{\mathfrak{m}}^{C_L}(\vec{x}) = \sum_{i=1}^{n}(\max\{\min\{x_{(i)}, \frac{\mathfrak{m}(A_{(i)})}{2}\}, x_{(i)} + \mathfrak{m}(A_{(i)}) - 1\} - \max\{\min\{x_{(i-1)}, \frac{\mathfrak{m}(A_{(i)})}{2}\}, x_{(i-1)} + \mathfrak{m}(A_{(i)}) - 1\})$
CDiv	$\mathfrak{C}_{\mathfrak{m}}^{C_{Div}}(\vec{x}) = \sum_{i=1}^{n}\left(\frac{x_{(i)}\mathfrak{m}(A_{(i)})+\min\{x_{(i)}, \mathfrak{m}(A_{(i)})\}}{2} - \frac{x_{(i-1)}\mathfrak{m}(A_{(i)})+\min\{x_{(i-1)}, \mathfrak{m}(A_{(i)})\}}{2}\right)$

- $y_p \in \mathbb{C} = \{C_1, C_2, \ldots, C_M\}$ is the label of the class of the pth training example.

Although one may find in the literature several techniques to deal with classification problems, FRBCSs [39] are indeed the one most frequently adopted. Observe that FRBCSs allow us to have all the available information in the system modeling, obtaining an interpretable model and producing quite accurate results. A FRBCSs is composed by:

(i) The Knowledge Base: it contains the Rule and the Data Bases, with, respectively, the fuzzy inference rules and the membership functions;
(ii) The Fuzzy Reasoning Algorithm: it is the inference procedure used to classify examples considering all the information stored in the Knowledge Base.

We adopt the following structure for the fuzzy rules:

$$\text{Rule } R_j: \text{ If } x_{p1} \text{ is } A_{j1} \text{ and } \ldots \text{ and } x_{pn} \text{ is } A_{jn} \text{ then Class is } C_j \text{ with } RW_j, \quad (7)$$

where:

- $x_p = (x_{p1}, \ldots, x_{pn})$ is the n-dimensional vector of attribute values corresponding to an example x_p of the pth example;
- R_j is the label of the jth rule;
- A_{ji} is an antecedent fuzzy set which models a linguistic term;
- C_j is the class of the jth rule;
- $RW_j \in [0, 1]$ is the rule weight [40], which, in this case, is computed using the certainty factor.

We adopt the FARC-HD [38] fuzzy classifier (Fuzzy Association Rule-based Classification model for High Dimensional problems), to accomplish the fuzzy rule learning process. The learning process of FARC-HD is composed of the following three stages:

1) *Fuzzy association rule extraction for classification:* This step is aimed at generating fuzzy association rules from frequent itemsets. To do so, a search tree is constructed for each class computing the confidence and support of each item or itemset (in this algorithm an item is a linguistic label). We have to point out that the number of antecedents of the different fuzzy rules can be different and their maximum length is determined by the maximum depth allowed for the search tree.
2) *Candidate rule prescreening:* This phase uses an instance weighting scheme to preselect the most interesting rules and consequently, to decrease the computational cost of the evolutionary process.
3) *Genetic rule selection and lateral tuning:* The final stage of the method consists in the application of an evolutionary algorithm to perform both the lateral tuning of the fuzzy sets [41] and the selection of the best rules generated in the previous step.

4.2. The new fuzzy reasoning method

Once the fuzzy rules composing the system have been created, it is necessary a mechanism for classifying new examples. Specifically, let $x_p = (x_{p1}, \ldots, x_{pn})$ be a new example to be classified, L being the number of rules in the rule base and M being the number of classes of the problem. The new algorithm where our CC-functions are applied in the FRM is composed of the following steps:

Step 1 To compute the *matching degree*, that is, the strength of the activation of the if-part of the rules for the example x_p, which is computed using a t-norm T': $[0, 1]^n \to [0, 1]$:

$$\mu_{A_j}(x_p) = T'(\mu_{A_{j1}}(x_{p1}), \ldots, \mu_{A_{jn}}(x_{pn})), \text{ with } j = 1, \ldots, L. \quad (8)$$

Step 2 The *Association degree* is computed, that is, for the class of each rule, the matching degree is weighted with the corresponding rule weight, given by:

$$b_j^k(x_p) = \mu_{A_j}(x_p) \cdot RW_j^k, \text{ with } k = Class(R_j), j = 1, \ldots, L. \quad (9)$$

Step 3 The *example classification soundness degree for all classes* is calculated, applying the CC-functions (Eq. (6)) to combine the association degrees obtained in the previous step, as follows:

$$Y_k(x_p) = \mathfrak{C}_{\mathfrak{m}}^{C}\left(b_1^k(x_p), \ldots, b_L^k(x_p)\right), \text{ with } k = 1, \ldots, M, \quad (10)$$

where $\mathfrak{C}_{\mathfrak{m}}^{C}$ is the obtained CC-integral, for the copula C: $[0, 1]^2 \to [0, 1]$ and fuzzy measure $\mathfrak{m}: 2^N \to [0, 1]$ (Table 2). Since, whenever $b_j^k(x_p) = 0$, it holds that:

$$\mathfrak{C}_{\mathfrak{m}}^{C}(b_1^k(x_p), \ldots, b_L^k(x_p)) = \mathfrak{C}_{\mathfrak{m}}^{C}(b_1^k(x_p), \ldots, b_{j-1}^k(x_p), b_{j+1}^k(x_p), \ldots, b_L^k(x_p)),$$

then, for practical reasons, only those $b_j^k > 0$ are considered in Eq. (10).

Step 4 A *Classification* decision function $F: [0, 1]^M \to \{1, \ldots, M\}$ is applied over the example classification soundness degrees of all classes and thus, the class corresponding to the maximum soundness degree is determined by:

$$F(Y_1, \ldots, Y_M) = \min_{k=1\ldots M} k \text{ s.t. } Y_k = \max_{w=1,\ldots,M}(Y_w). \quad (11)$$

In practical applications, it is sufficient to consider

$$F(Y_1, \ldots, Y_M) = \arg\max_{k=1,\ldots,M}(Y_k). \quad (12)$$

Please cite this article as: G. Lucca et al., CC-integrals: Choquet-like Copula-based aggregation functions and its application in fuzzy rule-based classification systems, Knowledge-Based Systems (2016), http://dx.doi.org/10.1016/j.knosys.2016.12.004

Table 3
Association degrees for each class.

	C_1	C_2	C_3
R_x	0.9	0.89	0.1
R_y	0.2	0.88	0.3
R_z	0.1	0.86	0.25

As it can be observed, in the third step of the FRM we propose to use CC-integrals, which are associated with a fuzzy measure. According to the results obtained in several works (see, e.g., in [12,15,36]), we have selected the power measure where the exponent q is genetically learned. Therefore, for each class of the problem we create a different fuzzy measure by learning the most appropriate exponent to model the interaction among the rules of that class. To do so, the CHC evolutionary algorithm [14] is considered since it is widely used in this domain [24,42]. The main specific features of our evolutionary model are the following ones.[3]

- **Coding Scheme:** A chromosome is formed of as many genes as classes (G_k, $k = 1, \ldots, M$), using a real codification. That is:

  $C_{CHOQUET} = \{G_1, \ldots, G_M\}$

  where $G_k \in [0.01, 1.99]$ with $k = 1, \ldots, M$. However, the suggested range of the real values the exponents, q_k, is [0.01, 100]. For this reason, it is necessary to adapt the value of the genes to the real range, as following:

$$q_k = \begin{cases} G_k & \text{if } 0 < G_k \leq 1 \\ \frac{1}{2-G_k} & \text{if } 1 < G_k < 2 \end{cases} \quad (13)$$

- **Initial Gene Pool:** An individual containing all genes with value 1 is included in the population. In this manner, the cardinality measure is obtained.
- **Chromosome Evaluation:** The most common metric for classification problems is used, i.e. the accuracy rate. This metric is the percentage of correctly classified examples.
- **Crossover Operator:** The parent centrix BLX crossover operator [43] is selected as it is usually applied for real codifications.
- **Restarting Approach** In order to avoid the local optima, the algorithm uses a restarting approach since it does not apply mutation during the recombination phase. To do so, a threshold is firstly initialized and it is decreased when no new individuals are included in the population after the crossover stage. When the threshold value is less than zero, all chromosomes (except the best one, as in the elitist scheme) are regenerated randomly with the purpose of introducing new diversity to the search. The evolutionary process also finishes when three restarts are sequentially done without improving the best solution.

In order to clarify how the new FRM works, we introduce a short example considering different aggregation functions to perform the third step of the FRM where the local information given by several rules (obtained after step 2 of the FRM) is aggregated according to the classes of the rules. More specifically, we compare the FRM that uses the maximum as aggregation function and our new methodology considering the CC-integral based on the minimum t-norm, *TM*.

Example 1. A classification problem composed of 3 classes is studied. In Table 3 we introduce the association degrees obtained by the fuzzy rules fired when classifying a new example. As it can be observed there are three fired rules for each class (columns of Table 3) and consequently, three aggregations have to be computed (one for each class) in the third step of the FRM.

[3] For more details about the evolutionary method see [12].

In order to show the behavior of our new method we consider the usage of the CC-integral based on the minimum t-norm (T_M) and the cardinality measure, that is, the power measure setting the exponent q to 1 (in order to ease the understanding of the calculations). We compare it versus the classical FRM of the winning rule, which uses the maximum as the aggregation function. The values computed for each class using the two approaches are the following ones:

- C_1
  - Sorted association degrees: [0.1, 0.2, 0.9]
    - Maximum = 0.9
    - TM = $(\min(0.1, \frac{3}{3}) - \min(0, \frac{3}{3})) + (\min(0.2, \frac{2}{3}) - \min(0.1, \frac{2}{3})) + (\min(0.9, \frac{1}{3}) - \min(0.2, \frac{1}{3})) = 0.33$
- C_2
  - Sorted association degrees: [0.86, 0.88, 0.89]
    - Maximum = 0.89
    - TM = $(\min(0.86, \frac{3}{3}) - \min(0, \frac{3}{3})) + (\min(0.88, \frac{2}{3}) - \min(0.86, \frac{2}{3})) + \min((0.89, \frac{1}{3}) - \min(0.88, \frac{1}{3})) = 0.86$
- C_3
  - Sorted association degrees: [0.1, 0.25, 0.3]
    - Maximum = 0.3
    - TM = $(\min(0.1, \frac{3}{3}) - \min(0, \frac{3}{3})) + (\min(0.25, \frac{2}{3}) - \min(0.1, \frac{2}{3})) + (\min(0.3, \frac{1}{3}) - \min(0.25, \frac{1}{3})) = 0.3$

Once the association degree for each class has been computed, the predicted class is the one associated with the largest value (last step of the FRM):

- Maximum = Max[**0.9**, 0.89, 0.3] = C_1
- TM = Max[0.33, **0.86**, 0.3] = C_2

It can be noticed that the usage of the maximum as an aggregation operator predicts class 1, since it only considers the information provided by one rule per class. However, if we look in detail at the association degrees presented in Table 3, this prediction may not be ideal, since that the class 1 has one rule having a high compatibility whereas the class 2 has three rules having high compatibilities (slightly less than that of class 1). Therefore, class 2 seems to be a most appropriate option. This fact is taken into account by our new approach since the information given by all the fuzzy rules and not only by the best one is considered and consequently, class 2 is predicted.

Furthermore, in the example it can be observed that the behavior of the CC-integrals can be different depending on the values to be aggregated. For example, for class 2 the result is the minimum of the values to be aggregated, for class 3 is the maximum whereas for class 1 is an intermediate value. Therefore, the FRM is more flexible, which may imply a enhancement of its behavior.

5. Experiment specifications

In this section, firstly we present the real world classification problems besides the configuration for the considered approaches. After that, we introduce the statistical tests that are necessary to compare the achieved results.

5.1. Datasets

We have selected 30 real world datasets from the KEEL dataset repository [23]. Table 4 presents the characteristics of the these datasets: the identifier (Id.), the name (Dataset), the number of instances (*#Inst*), the number of attributes (*#Att*) and the number of classes (*#Class*).

Some datasets, namely: *magic, page-blocks, penbased, ring, satimage* and *twonorm*, were stratified sampled at 10% in order to reduce their size for training. Some examples containing missing information were removed, e.g., in the *wisconsin* dataset.

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Table 4
Summary of the properties of the considered datasets.

Id.	Dataset	#Inst	#Att	#Class
App	Appendiciticis	106	7	2
Bal	Balance	625	4	3
Ban	Banana	5300	2	2
Bnd	Bands	365	19	2
Bup	Bupa	345	6	2
Cle	Cleveland	297	13	5
Eco	Ecoli	336	7	8
Gla	Glass	214	9	6
Hab	Haberman	306	3	2
Hay	Hayes-Roth	160	4	3
Iri	Iris	150	4	3
Mag	Magic	1,902	10	2
New	Newthyroid	215	5	3
Pag	Pageblocks	5,472	10	5
Pen	Penbased	10,992	16	10
Pho	Phoneme	5,404	5	2
Pim	Pima	768	8	2
Rin	Ring	740	20	2
Sah	Saheart	462	9	2
Sat	Satimage	6,435	36	7
Seg	Segment	2,310	19	7
Shu	Shuttle	58,000	9	7
Spe	Spectfheart	267	44	2
Tit	Titanic	2,201	3	2
Two	Twonorm	740	20	2
Veh	Vehicle	846	18	4
Vow	Vowel	990	13	11
Win	Wine	178	13	3
Wis	Wisconsin	683	11	2
Yea	Yeast	1,484	8	10

The model used in this paper, which was proposed in [12,36,42], consists of a 5-fold cross-validation model, where a dataset is split in five partitions randomly. Each partition presents 20% of the examples. Four partitions are used for training, and the other is used for testing. This process is repeated five times, using a different partition to test the formed system each time. The performance of the approaches are measured considering each partition, based on the accuracy rate, which is defined by the number of correctly classified examples divided by the total number of examples for each partition. After, is calculate the average result of the five testing partitions, which is output of the algorithm.

5.2. Configuration of the different classifiers used in the study

The parameter set-up for the FARC-HD algorithm is the one suggested by the authors:

- The conjunction operator T' is the product t-norm;
- The rule weight RW_j is the certainty factor;
- 5 linguistic labels per variable;
- The minimum support is 0.05;
- The threshold for the confidence is 0.8;
- The depth of the search trees is limited to 3;
- k_t, the parameter that determines the number of fuzzy rules that cover each example, is equal to 2.

The features considered for the evolutionary process are the following ones:

- The populations are composed of 50 individuals;
- 30 bits per gen are considered for the Gray codification;
- The maximum number of evaluations is 20.000.

Finally, for the copula O_α, the value adopted for the α parameter is 0.1 ($\alpha = 0.1$), since this value was tested in [36] previously, although in a different method, but providing better performance than the other values.

5.3. Statistical tests for performance comparison

In order to give statistical support for the analysis of the results, we consider the usage of hypothesis validation techniques [24,25]. Specifically, we use non-parametric tests, since the initial conditions that guarantee the reliability of the parametric tests cannot be performed [26].

In fact, we use the aligned Friedman rank test [44] to detect statistical differences among a group of results and to show how good a method is with respect to its partners. In this method, the algorithm achieving the lowest average ranking is the best one. Additionally, we have graphically shown the obtained ranks to easily observe which is the best method.

Furthermore, we apply the post-hoc Holm's test [45] to study whether the best ranking method rejects the equality hypothesis with respect to its partners. The post-hoc procedure allows us to know if a hypothesis of comparison could be rejected at a specified level of significance α. Specifically, we compute the adjusted p-value (APV) to take into account that multiple tests are conducted. As a result, we can directly compare the APV with the level of significance α so as to be able to reject the null hypothesis.

Finally, we also consider the usage of the Wilcoxon test [46] in order to perform pair-wise comparisons.

6. Analysis of the application of the CC-integral in classification problems

This section is aimed at providing an analysis of the application of CC-integral in real-world classification problems. As we have mentioned in Section 4, we use the FARC-HD fuzzy classifier whose FRM is adapted to use CC-integrals (the ones introduced in Table 2). Specifically, the goal of the analysis is to study whether the usage of CC-integrals in the FRM allows us to obtain results as accurate as those achieved by the best performing pre-aggregation functions introduced in [15], which is named in this paper Ham_{PA}.

To do so, the study is divided in three main parts:

- In first place, we want to determine the best CC-Integral. For that, we conduct a study comparing the different copulas considered in this work (the ones shown in Table 1), which produce all the CC-integrals presented in Table 2. For this task, we divide the study according to the type of copula: t-norm, overlap or copula (which is neither a t-norm nor an overlap function), and finally, we compare the best of each group to determine the best CC-integral (Section 6.1).
- Then, we compare if the best CC-integral is able to improve the results provided by the classical FRM of the WR as well as the standard Choquet integral and those provided by the best pre-aggregation function introduced [15], that is, the Ham_{PA} (Section 6.2).
- Finally, we analyze the execution time taken by the different approaches (Section 6.3).

6.1. Analyzing the behavior of the different CC-integrals

This subsection is aimed at analyzing the behavior of the proposed generalizations in the FRM. In order to determine the best CC-integral in this study we compare them by groups of copulas according to their types (copulas that are t-norms, overlaps and those that are neither t-norms nor overlap functions, the last denote by n-copulas). From each group, we select the best one and we compare them to determine the best CC-integral.

The results achieved in testing by the different CC-integrals are presented in Table 5 by columns, and in each line, containing the mean of accuracy obtained in the five partitions of the dataset where the best global result for each dataset is highlighted in **boldface**.

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Table 5
Accuracy rate achieved in testing for the different CC-integrals.

Dataset	TM	TL	TH	OB	Oα	OmM	CF	CL	Cdiv
App	**85.84**	82.99	84.94	81.13	**85.84**	81.13	83.98	83.03	83.03
Bal	81.60	80.64	80.96	81.28	79.52	80.32	81.28	**82.24**	78.88
Ban	84.30	84.43	85.91	86.17	86.11	**87.00**	85.85	86.79	85.38
Bnd	**71.06**	68.24	69.38	66.01	67.72	67.41	70.47	68.54	67.70
Bup	61.45	65.51	64.06	63.19	**67.25**	66.38	65.51	62.90	64.93
Cle	54.88	**57.57**	56.91	56.92	55.23	56.55	55.92	54.54	54.89
Eco	77.09	73.53	76.20	74.41	**78.86**	76.50	77.40	78.58	77.38
Gla	**69.17**	61.71	64.97	66.39	63.55	60.29	60.76	63.12	66.82
Hab	74.17	73.21	70.90	71.22	73.51	69.93	**74.48**	72.86	72.86
Hay	**81.74**	79.46	79.49	78.75	79.49	78.72	78.75	79.49	78.75
Iri	92.67	93.33	93.33	93.33	92.67	**94.67**	94.00	92.67	93.33
Mag	79.81	**80.34**	79.86	79.34	80.07	79.02	80.02	79.97	79.07
New	93.95	93.02	93.95	94.42	94.42	**96.28**	94.42	93.95	94.88
Pag	93.97	93.98	93.61	93.97	**94.88**	94.16	94.16	93.79	94.34
Pen	91.27	90.82	**91.91**	91.45	90.09	89.82	91.36	90.91	90.73
Pho	82.94	81.83	83.40	**83.55**	83.46	82.62	82.73	83.53	83.10
Pim	75.78	74.87	74.87	75.38	75.78	73.57	74.86	76.04	**76.95**
Rin	87.97	89.46	89.19	89.73	**90.41**	88.38	89.59	88.65	89.59
Sah	**70.78**	69.03	69.48	67.53	66.89	67.95	68.83	68.82	69.69
Sat	79.01	78.54	**79.94**	79.62	78.85	78.69	79.63	79.16	79.00
Seg	92.25	91.90	93.12	92.68	92.86	92.77	92.55	**93.20**	92.94
Shu	**98.16**	97.66	97.29	97.52	97.47	97.24	96.41	97.79	98.07
Spe	78.99	78.27	79.73	77.13	**79.74**	77.51	79.00	77.86	77.88
Tit	**78.87**	**78.87**	**78.87**	**78.87**	**78.87**	**78.87**	**78.87**	**78.87**	**78.87**
Two	85.14	**85.68**	84.59	84.73	84.59	85.14	84.32	85.14	83.92
Veh	**69.86**	66.67	67.97	67.85	67.73	68.32	68.56	67.85	68.56
Vow	**68.89**	66.26	67.88	66.57	67.47	67.37	66.97	65.76	68.38
Win	93.83	95.48	96.08	94.35	94.38	96.62	96.62	**98.32**	95.51
Wis	95.90	96.34	96.63	96.34	**97.22**	96.34	97.22	96.34	96.63
Yea	57.01	57.75	57.61	57.28	56.40	57.34	57.55	57.82	**57.88**
Mean	**80.28**	79.58	80.10	79.57	80.04	79.56	80.07	79.95	80.00

Table 6
Aligned Friedman and Holm test to compare the different CC-integrals derived from different families of copulas.

t-Norm	TM	TL	TH
	39.61	56.31 (0.02)	40.56 (0.88)
Overlap	OB	O_α	OmM
	47.63 (0.23)	**39.08**	47.78 (0.23)
n-Copula	CL	CF	Cdiv
	39.76 (0.33)	**49.08**	47.65 (0.33)

From the results shown in the previous table, it is possible notice that the behavior of the different CC-integrals derived from the same family of copulas are quite similar among themselves with the exception of Lukasiewicz (for t-norms) and OmM (for overlaps), which provided the worst results. Furthermore, it seems that the CC-integrals associated with the minimum t-norm, TM, achieved the best result in 9 datasets and the best average result among all CC-integrals under study. Therefore, a priori, TM would be the best CC-integral. Anyway, these conclusions cannot be meaningful without conducting the appropriate study.

Specifically, we have carried out three multiple comparison statistical tests (one for each family of aggregation functions) by applying the aligned Friedman test and the post-hoc Holm's test, whose results are shown in Table 6. These results are grouped by rows according to the family of aggregation functions. In each row, we show the results of the three functions considered in this paper, for each family. The value of each cell correspond to the rank obtained by the aligned Friedman test, when comparing the different functions belonging to the same family. The value shown in brackets represent the APV obtained by the Holm post-hoc test, using as the control method the approach having the smallest rank, which is shown in **boldface**. The APV is underlined when there are statistical differences (considering a significance level of $\alpha = 0.1$).

As is shown in the first row of Table 6, when considering t-norms, the results provided by TM are statistically better than those by TL and similar to those by TH. However, the fact that TM obtained a better overall mean providing the best result in 12 datasets leads us to select it as the best t-norm.

Regarding CC-integrals derived from overlap functions, we can see in the second row of Table 6 that the best ranked one is O_α, but there are not statistical differences among them. From this group, we select O_α as the representative, since it obtained the best overall mean and achieved the highest accuracy rate in 14 data-sets and tie only once.

Finally, the third row of Table 6 allows us to state that CF is the best option among the group of n-copulas. Although there are not statistical differences among the functions in this family, this copula obtained the best ranking, which is based on obtaining the best result in 11 out of the 30 datasets, having a tie in two datasets.

Once the best aggregation is determined for each family of copulas, namely, TM, Oα and CF, we conduct another statistical study to compare them among themselves. We have carried out again the aligned Friedman test to detect whether there are statistical differences among them or not. The obtained rankings are shown in Fig. 1. Then we run the Holm's statistical test using TM as control method, since it provides the best ranking in the previous statistical test. The obtained results by the Holm's test are shown in Table 7. From these results we can conclude that TH does not statistically outperform neither O_α nor CF. However, we select TM as the best option due to the fact that it obtained the best overall result, as shown in Table 5.

6.2. Comparison of the best CC-integral versus other averaging operators

As we have shown in the previous section, the best CC-integral among the ones considered in this is study is the one related to the minimum t-norm, TM. In this section our aim is to compare this generalization against the classical FRM of the WR (it uses the maximum as aggregation function), the standard Choquet In-

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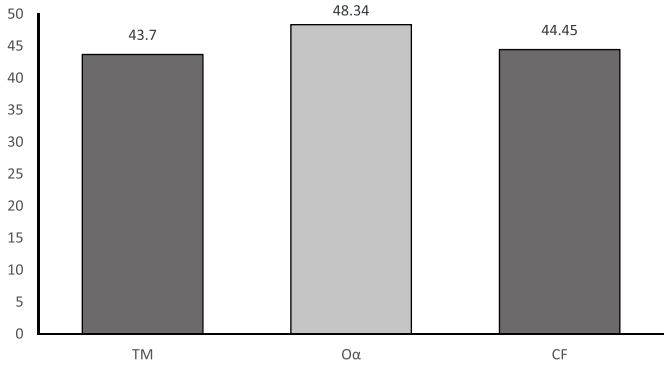


Fig. 1. Rankings obtained by the best CC-integrals for each family of copulas.

Table 7
Holm test to compare the best CC-integrals for each family of copulas.

Generalization	APV
Oα	0.98
CF	0.98

Table 8
Accuracy results achieved in test provided by the best CC-integral and other averaging operators.

Dataset	TM	Choquet	Ham_{PA}	WR
App	**85.84**	80.13	82.99	83.03
Bal	81.60	82.40	**82.72**	81.92
Ban	84.30	**86.32**	85.96	83.94
Bnd	71.06	68.56	**72.13**	69.40
Bup	61.45	**66.96**	65.80	62.03
Cle	54.88	55.58	55.58	**56.91**
Eco	77.09	76.51	**80.07**	75.62
Gla	**69.17**	64.02	63.10	64.99
Hab	**74.17**	72.52	72.21	70.89
Hay	**81.74**	79.49	79.49	78.69
Iri	92.67	91.33	93.33	**94.00**
Mag	**79.81**	78.86	79.76	78.60
New	93.95	94.88	**95.35**	94.88
Pag	93.97	94.16	**94.34**	94.16
Pen	91.27	90.55	90.82	**91.45**
Pho	82.94	82.98	**83.83**	82.29
Pim	**75.78**	74.60	73.44	74.60
Rin	87.97	**90.95**	88.78	90.00
Sah	**70.78**	69.69	70.77	68.61
Sat	79.01	79.47	**80.40**	79.63
Seg	92.25	**93.46**	93.33	93.03
Shu	**98.16**	97.61	97.20	96.00
Spe	**78.99**	77.88	76.02	77.90
Tit	**78.87**	**78.87**	**78.87**	**78.87**
Two	85.14	84.46	85.27	**86.49**
Veh	**69.86**	68.44	68.20	66.67
Vow	**68.89**	67.58	68.18	67.98
Win	93.83	93.79	**96.63**	96.60
Wis	95.90	**97.22**	96.78	96.34
Yea	**57.01**	55.73	56.53	55.32
Mean	**80.28**	79.83	80.26	79.70

tegral (the one that uses the product t-norm) and the best pre-aggregation function from [15] (Ham_{PA}). Table 8 shows the results achieved in testing by these four approaches, where the best global result for each dataset is highlighted in **boldface**.

Looking at the results of these four approaches, it is noticeable that TM obtained the best average global result, which is similar to that of Ham_{PA} and it is superior to those of WR and the standard Choquet integral. In a closer look we can observe that TM obtains the best result in 12 datasets, whereas Ham_{PA}, Choquet and WR provide the best results in 8, 5 and 4 datasets, respectively.[4] In order to support our previous results, we have carried out a set of pair-wise statistical comparisons using the well-known Wilcoxon signed-rank test [46]. Specifically, we have compared the best CC-integral, TM, versus WR, Ham_{PA} and Choquet integral. Table 9 shows the results of these comparisons, where R^+ indicates the ranks obtained by TM and R^- represents the ranks achieved by the method used in each comparison.

Table 9
Wilcoxon test to compare the best CC-integral versus the Ham_{PA}, the Winning Rule and the standard Choquet integral.

Comparison	R^+	R^-	p-value
TM vs. Ham_{PA}	215.0	250.0	0.72
TM vs. WR	311.5	153.5	0.09
TM vs. Choquet	303.5	161.5	0.14

According to the obtained statistical results presented in Table 9, we can affirm, with a high level of confidence, that the CC-integral defined by the minimum is better than the WR. Regarding the standard Choquet integral, we can observe that, although there are not statistical differences, the obtained p-value is low. Furthermore, TM improves the results of the Choquet integral in 18 out of the 30 datasets considered in this study. These two facts, show that TM is enhancing the results provided by the standard Choquet integral. Finally, we point out that, when comparing TM and Ham_{PA}, the obtained p-value is high, which implies that the behavior of these two approaches is similar.

6.3. Analyzing the execution time of the different CC-integrals

In this subsection we present the analysis of the results in terms of the execution time. We must point out that this execution time refers to all process, including the reading of the files containing the datasets, the execution of the learning algorithm as well as the classification of the training and testing examples and the writing of the output files.

The experimnets have been carried out in a 8 nodes cluster connected via 1GB/s Ethernet LAN network. Half of these nodes are composed by 2 Intel Xeon E5-2620 v3 processors at 2.4 GHz (3.2 GHz with Turbo Boost) with 12 virtual cores in each one (where 6 of them are physical). Three of the remaining node are equipped with 2 Intel Xeon E5-2620 v2 processors at 2.1 GHz with the same number of cores than the previous ones. The last node is the master node, composed of an Intel Xeon E5-2609 processors with 4 physical cores at 2.4 GHz. All slaves nodes are equipped with 32GB of RAM memory, while the master works with 8GB of RAM memory. With the respect to storage specifications, all nodes uses Hard Disk Drivers featuring read/write performance of 128 MB/s.

We present in Table 10 the mean execution time of all different approaches considered in the study and we highlight in **boldface** the aggregation that achieves the lowest execution time. From the execution times achieved, it is noticeable that all CC-integrals have a similar run-time (more or less one minute), being TL the quickest. In relation to the standard Choquet integral and the pre-aggregation function that considers the Hamacher product (Ham_{PA}) it is possible to observe that they have a similar execution time, even similar to the CC-integrals. This behavior is normal since all of them are based on the Choquet integral and the number of computations is the same. The small differences are due to the different aggregation function used.

Regarding the FRM of the WR, as expected, it is the approach having the less execution time. This occurs because in this FRM it

[4] We have not considered in the count the Titanic dataset which have the same classification rate for all methods.

Table 10
Average execution time considering all aggregations.

Dataset	TM	TL	TH	OB	Oα	OmM	CF	CL	CDiv	Choquet	Ham$_{PA}$	WR
App	0:05	0:05	0:05	0:04	0:06	0:06	0:06	0:06	0:05	0:05	0:05	**0:01**
Bal	0:57	0:57	0:59	0:59	0:57	0:48	0:47	0:42	0:41	0:54	1:09	**0:23**
Ban	2:37	2:16	2:36	2:32	2:15	3:40	3:13	2:11	2:50	2:41	2:19	**0:25**
Bnd	0:28	0:27	0:33	0:34	0:32	0:32	0:29	0:25	0:29	0:31	0:29	**0:14**
Bup	0:19	0:15	0:18	0:19	0:19	0:20	0:17	0:16	0:19	0:19	0:18	**0:03**
Cle	0:29	0:25	0:30	0:32	0:33	0:32	0:34	0:30	0:28	0:35	0:31	**0:19**
Eco	0:21	0:17	0:20	0:22	0:22	0:20	0:22	0:22	0:23	0:22	0:24	**0:09**
Gla	0:11	0:10	0:13	0:14	0:14	0:13	0:14	0:11	0:12	0:12	0:13	**0:05**
Hab	0:12	0:11	0:12	0:12	0:14	0:12	0:12	0:12	0:13	0:12	0:11	**0:02**
Hay	0:05	0:05	0:05	0:05	0:06	0:05	0:05	0:04	0:04	0:05	0:05	**0:03**
Iri	0:02	0:02	0:02	0:02	0:03	0:02	0:02	0:03	0:02	0:02	0:03	**0:00**
Mag	2:22	2:41	2:50	2:39	2:41	4:29	3:41	2:23	2:27	2:47	2:54	**0:55**
New	0:07	0:06	0:08	0:07	0:09	0:07	0:07	0:07	0:07	0:08	0:09	**0:02**
Pag	0:23	0:20	0:23	0:22	0:24	0:26	0:23	0:25	0:23	0:23	0:22	**0:09**
Pen	2:07	1:28	1:44	1:41	1:40	1:29	1:37	1:26	1:37	2:10	1:48	**1:14**
Pho	3:35	3:36	4:04	3:58	5:27	3:46	3:26	2:57	3:07	5:04	4:57	**1:11**
Pim	1:27	1:23	1:20	1:25	1:32	2:03	1:39	1:23	1:18	1:22	1:09	**0:27**
Rin	0:41	0:48	0:40	0:42	0:32	0:40	0:37	0:31	0:29	0:46	0:33	**0:23**
Sah	0:44	0:38	0:43	0:40	0:36	0:36	0:37	0:33	0:33	0:48	0:35	**0:18**
Sat	1:17	1:03	1:01	0:58	1:00	0:52	0:52	**0:48**	0:50	1:14	1:04	0:57
Seg	3:15	**1:56**	2:46	2:50	3:50	2:36	2:34	2:36	2:38	3:56	3:52	1:57
Shu	1:04	0:48	1:00	0:50	1:04	0:48	0:44	0:58	0:57	1:15	0:54	**0:18**
Spe	0:36	0:33	0:34	0:31	0:35	0:28	0:28	**0:26**	0:27	0:37	0:29	0:27
Tit	0:51	0:50	0:53	0:51	1:02	0:44	0:42	0:39	0:39	0:52	1:02	**0:10**
Two	0:45	0:46	0:43	0:45	0:47	0:59	0:54	0:42	0:44	0:44	0:44	**0:29**
Veh	2:07	1:32	1:50	1:40	1:32	1:30	1:36	1:26	1:25	1:52	1:42	**1:07**
Vow	1:55	1:16	1:56	1:55	1:31	1:56	2:07	1:47	1:51	1:53	1:39	**0:49**
Win	0:02	0:02	0:02	0:02	0:02	**0:01**	0:02	0:02	0:03	0:02	0:02	**0:01**
Wis	0:30	0:29	0:28	0:32	0:21	0:36	0:33	0:28	0:24	0:33	0:27	**0:10**
Yea	1:49	1:38	1:57	1:57	2:22	2:15	2:30	1:41	1:47	2:04	2:17	**0:46**
Mean	01:03	0:54	1:02	1:01	1:05	1:06	1:03	0:53	0:55	1:09	01:05	**0:27**

is not necessary to perform as many calculations as with the CC-integrals based FRM (see Example 1).

7. Conclusion

In this paper, we introduce the notion of Choquet-like Copula-based aggregation function (CC-integral). We applied the CC-integral based on several kinds of copulas in FRBCSs, showing that the one based on the minimum t-norm presented the best results among the CC-integrals considered in this work. Furthermore, we highlight that this CC-integral allows to enhance the results of the classical FRM of the WR as well as those of the standard Choquet integral, and provides results that are competitive with those obtained by the Ham$_{PA}$, offering new possibilities in defining FRMs with similar gain in performance.

In future works, we intend to study the properties satisfied by the CC-integrals. We will also consider CC-integral in a fuzzy interval approach [47–51], as, e.g., in [52,53].

Acknowledgment

This work is supported by Brazilian National Counsel of Technological and Scientific Development CNPq (Proc. 232827/2014-1, 233950/2014-1, 481283/2013-7, 306970/2013-9, 307681/2012-2) and by the Spanish Ministry of Science and Technology under project TIN2016-77356-P

Appendix A. Abbreviations used in the paper

In order to make a paper easy to read and clearer, we present in this appendix a table containing the abbreviations used throughout the paper. In Table A.11 is presented in each line the acronym followed by its meaning.

Table A.11
Abbreviations used in the paper.

Abbreviation	Meaning
CC-integral	Choquet-Like Copula-Based integral
t-norm	Triangular norm
WR	Winning Rule
FRBCSs	Fuzzy Rule-Based Classification Systems
FRM	Fuzzy Reasoning Method
CHC	Crossover using generational elitist selection with Heterogeneous recombination and Cataclysm mutation
FARC-HD	Fuzzy Association Rule-based Classification model for High Dimensional problems

Appendix B. Example of an overlap function that is a non-associative copula

We show that the overlap function O_B is a copula. Consider x, y, x', $y' \in [0, 1]$ such that $x \leq x'$ and $y \leq y'$. First observe that an equivalent definition for O_B is the following one: $O_B(x, y) = \min\{x, y\}\sqrt{\max\{x, y\}}$. Since O_B is commutative, then it is sufficient to consider the following three cases:

Case 1: $x \leq x' \leq y \leq y'$. In this case, one has that

$$O_B(x, y) = x\sqrt{y},$$
$$O_B(x, y') = x\sqrt{y'},$$
$$O_B(x', y) = x'\sqrt{y},$$
$$O_B(x', y') = x'\sqrt{y'}.$$

So, it holds that

$$O_B(x', y') - O_B(x', y) = x'\sqrt{y'} - x'\sqrt{y} = x'(\sqrt{y'} - \sqrt{y}),$$
$$O_B(x, y') - O_B(x, y) = x\sqrt{y'} - x\sqrt{y} = x(\sqrt{y'} - \sqrt{y}).$$

Since $x \leq x'$, then one concludes that

$$O_B(x', y') - O_B(x', y) \geq O_B(x, y') - O_B(x, y).$$

Please cite this article as: G. Lucca et al., CC-integrals: Choquet-like Copula-based aggregation functions and its application in fuzzy rule-based classification systems, Knowledge-Based Systems (2016), http://dx.doi.org/10.1016/j.knosys.2016.12.004

Case 2: $x \le y \le x' \le y'$. In this case, one has that

$$O_B(x,y) = x\sqrt{y},$$
$$O_B(x,y') = x\sqrt{y'},$$
$$O_B(x',y) = y\sqrt{x'},$$
$$O_B(x',y') = x'\sqrt{y'}.$$

So, it holds that

$$O_B(x',y') - O_B(x,y') = x'\sqrt{y'} - x\sqrt{y'} = \sqrt{y'}(x'-x),$$

and

$$\begin{aligned} O_B(x',y) - O_B(x,y) &= y\sqrt{x'} - x\sqrt{y} \\ &= \sqrt{y}(\sqrt{y}\sqrt{x'} - x) \\ &= \sqrt{y}(\sqrt{yx'} - x). \end{aligned}$$

Since $\sqrt{y} \le \sqrt{y'}$ and $\sqrt{yx'} \le \sqrt{x'x'} = x'$, then it follows that

$$O_B(x',y') - O_B(x,y') \ge O_B(x',y) - O_B(x,y).$$

Case 3: $x \le y \le y' \le x'$. In this case, one has that

$$O_B(x,y) = x\sqrt{y},$$
$$O_B(x,y') = x\sqrt{y'},$$
$$O_B(x',y) = y\sqrt{x'},$$
$$O_B(x',y') = y'\sqrt{x'}.$$

So, it holds that

$$O_B(x',y') - O_B(x',y) = y'\sqrt{x'} - y\sqrt{x'} = \sqrt{x'}(y'-y),$$
$$O_B(x,y') - O_B(x,y) = x\sqrt{y'} - x\sqrt{y} = x(\sqrt{y'} - \sqrt{y}).$$

Since $x \le x'$ and $(\sqrt{y'} - \sqrt{y}) \le y' - y$, then one concludes that

$$O_B(x',y') - O_B(x',y) \ge O_B(x,y') - O_B(x,y).$$

Therefore, in all cases, $O_B(x',y') + O_B(x,y) \ge O_B(x,y') + O_B(x',y)$, that is, O_B satisfy **(C1)**. Since **(C2)** and **(C3)** are trivially satisfied by O_B then O_B is non-associative copula.

Appendix C. Proofs

This appendix is aimed to present the proof of Proposition 5, which is recalled below.

Proposition 5. *For any copula* C: $[0,1]^2 \to [0,1]$ *and fuzzy measure* $\mathfrak{m} : 2^N \to [0,1]$, $\mathfrak{C}^C_{\mathfrak{m}}$ *is increasing* **(A1)**.

Proof. Since $\mathfrak{C}^C_{\mathfrak{m}}$ is trivially commutative, then it is sufficient to consider the case when the input $\vec{x}$ is ordered, that is, $x_i = x_{(i)}$, for each $i = 1, \ldots, n$. Also, by the transitivity, is it is sufficient to consider the following cases:

(i) Consider $x_{(j)} \le y \le x_{(j+1)}$, for some $j = 1, \ldots, n-1$. Since, C is a copula, then it holds that:

$$\begin{aligned} &C(x_{(j)}, \mathfrak{m}(A_{(j+1)})) + C(y, \mathfrak{m}(A_{(j)})) \\ &\quad \ge C(x_{(j)}, \mathfrak{m}(A_{(j)})) + C(y, \mathfrak{m}(A_{(j+1)})). \end{aligned}$$

Thus, one has that

$$\begin{aligned} &C(x_{(j)}, \mathfrak{m}(A_{(j)})) - C(x_{(j)}, \mathfrak{m}(A_{(j+1)})) \\ &\le C(y, \mathfrak{m}(A_{(j)})) - C(y, \mathfrak{m}(A_{(j+1)})) \end{aligned}$$

and, therefore, it holds that

$$\begin{aligned} &C(x_{(j)}, \mathfrak{m}(A_{(j)})) - C(x_{(j-1)}, \mathfrak{m}(A_{(j)})) + C(x_{(j+1)}, \mathfrak{m}(A_{(j+1)})) \\ &\quad - C(x_{(j)}, \mathfrak{m}(A_{(j+1)})) \\ &\le C(y, \mathfrak{m}(A_{(j)})) - C(x_{(j-1)}, \mathfrak{m}(A_{(j)})) + C(x_{(j+1)}, \mathfrak{m}(A_{(j+1)})) \\ &\quad - C(y, \mathfrak{m}(A_{(j+1)})). \end{aligned}$$

It follows that

$$\begin{aligned} &\mathfrak{C}^C_{\mathfrak{m}}(x_1, \ldots, x_j, \ldots, x_n) \\ &= \sum_{i=1}^{n} C(x_{(i)}, \mathfrak{m}(A_{(i)})) - C(x_{(i-1)}, \mathfrak{m}(A_{(i)})) \\ &\le \left(\sum_{i=1}^{j-1} C(x_{(i)}, \mathfrak{m}(A_{(i)})) - C(x_{(i-1)}, \mathfrak{m}(A_{(i)})) \right) \\ &\quad + \left(C(y, \mathfrak{m}(A_{(j)})) - C(x_{(j-1)}, \mathfrak{m}(A_{(j)})) \right) \\ &\quad + \left(C(x_{(j+1)}, \mathfrak{m}(A_{(j+1)})) - C(y, \mathfrak{m}(A_{(j+1)})) \right) \\ &\quad + \left(\sum_{i=j+2}^{n} C(x_{(i)}, \mathfrak{m}(A_{(i)})) - C(x_{(i-1)}, \mathfrak{m}(A_{(i)})) \right) \\ &= \mathfrak{C}^C_{\mathfrak{m}}(x_1, \ldots, x_{j-1}, y, x_{j+1}, \ldots, x_n). \end{aligned}$$

(ii) Consider $x_{(n)} \le y$. Since C is increasing, it holds that

$$C(x_{(n)}, \mathfrak{m}(A_{(n)})) \le C(y, \mathfrak{m}(A_{(n)})),$$

and, therefore, one has that

$$\begin{aligned} &C(x_{(n)}, \mathfrak{m}(A_{(n)})) - C(x_i, A_{(n-1)}, \mathfrak{m}(A_{(n)})) \\ &\le C(y, \mathfrak{m}(A_{(n)})) - C(x_i, A_{(n-1)}, \mathfrak{m}(A_{(n)})). \end{aligned}$$

It follows that:

$$\begin{aligned} \mathfrak{C}^C_{\mathfrak{m}}(x_1, \ldots, x_n) &= \sum_{i=1}^{n} C(x_{(i)}, \mathfrak{m}(A_{(i)})) - C(x_{(i-1)}, \mathfrak{m}(A_{(i)})) \\ &\le \left(\sum_{i=1}^{n-1} C(x_{(i)}, \mathfrak{m}(A_{(i)})) - C(x_{(i-1)}, \mathfrak{m}(A_{(i)})) \right) \\ &\quad + \left(C(y, \mathfrak{m}(A_{(n)})) - C(x_{(n-1)}, \mathfrak{m}(A_{(n)})) \right) \\ &= \mathfrak{C}^C_{\mathfrak{m}}(x_1, \ldots, x_{n-1}, y). \end{aligned}$$

□

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Please cite this article as: G. Lucca et al., CC-integrals: Choquet-like Copula-based aggregation functions and its application in fuzzy rule-based classification systems, Knowledge-Based Systems (2016), http://dx.doi.org/10.1016/j.knosys.2016.12.004

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Please cite this article as: G. Lucca et al., CC-integrals: Choquet-like Copula-based aggregation functions and its application in fuzzy rule-based classification systems, Knowledge-Based Systems (2016), http://dx.doi.org/10.1016/j.knosys.2016.12.004